Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 36)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the efficiency and cost of four types of solar photovoltaic (PV) cell.

Table 1

PV cell type	Efficiency / %	Relative cost per m^2
Amorphous silicon (thin film)	8	1
Polycrystalline silicon	16	2
Monocrystalline silicon with anti-reflective surface	22	3
Multi-junction cell	40	25

0 1 . 1 Describe how a photovoltaic cell generates electricity.

[2 marks]

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0 1 . 2 Explain how anti-reflective surfaces and multi-junction cells increase efficiency.

[3 marks]

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0 1 . 3 Use Table 1 to suggest which cell type would be chosen for a large solar farm on cheap desert land and which for a satellite. Justify your choices.

[3 marks]

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0 1 . 4 Explain why solar PV output is intermittent **and** unpredictable.

[2 marks]

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0 2 A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 2 . 1 Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 2 . 2 Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

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0 2 . 3 The test gave p = 0.02. Explain what this means.

[2 marks]

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0 2 . 4 Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

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0 2 . 5 Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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0 3 . 1 Explain how each of the following building design features reduces energy use.

[6 marks]

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Orientation

Low surface area : volume ratio

Earth sheltering

0 3 . 2 Compare the embodied energy of concrete with alternative building materials.

[2 marks]

0 3 . 3 Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

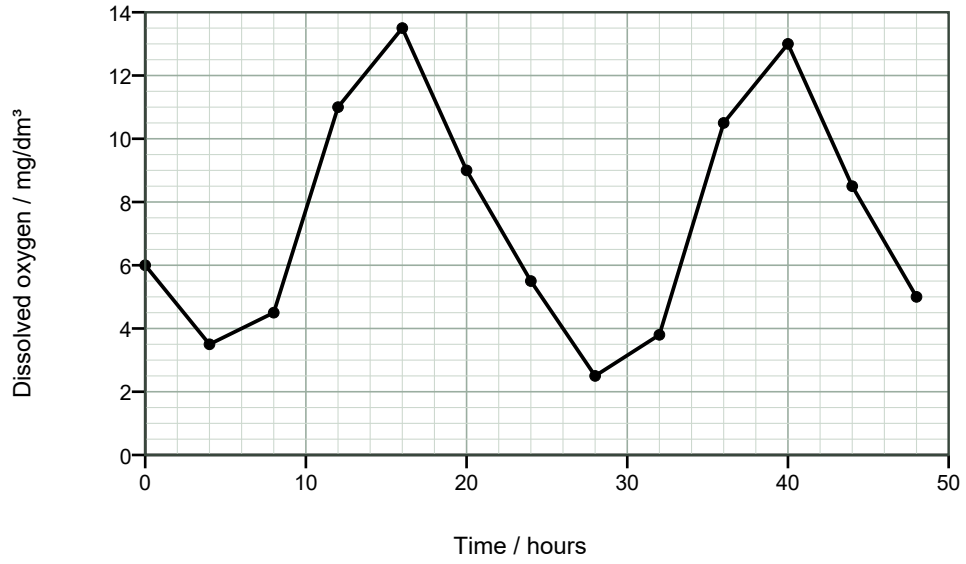
[2 marks]

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0 4 Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 4 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 4 . 2 Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 4 . 3 Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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0 4 . 4 Give **two** other harmful effects of algal blooms.

[2 marks]

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0 5 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 5 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 6 . 1 Explain why current global energy use is described as unsustainable.

[3 marks]

0 6 . 2 Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

0 6 . 3 Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

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0 7 Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

0 7 . 1 Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer ____ %

0 7 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

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Ultraviolet (UV) .....

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Visible light .....

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Infrared (IR) .....

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0 7 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

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0 7 . 4 Explain the term **thermal stratification** of the atmosphere.

[1 mark]

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0 8 . 1 Give **three** advantages of incinerating domestic waste.

[3 marks]

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0 8 . 2 Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

High combustion temperature with rapid cooling of gases .....

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Water spray scrubber .....

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Removal of wet / compostable waste .....

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Removal of recyclable waste .....

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0 8 . 3 Explain why the ash from waste incinerators needs careful disposal.

[2 marks]

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0 8 . 4 State **one** reason why incineration is expensive.

[1 mark]

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0 9 Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	**Metered households**
Mean water use / dm^3 per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 9 . 1 Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer ..... %

0 9 . 2 Explain why metering reduces water use.

[2 marks]

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0 9 . 3 Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

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0 9 . 4 Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 9 . 5 Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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1 0 . 1 Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

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1 0 . 2 State what is meant by **bioaccumulation**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS